

# Flying through climbable blocks in creative mode slows you down

MC-12829 Player

|          |                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------|
| Status   | Resolved / Fixed                                                                                                |
| Affected | Minecraft 1.5.1, Minecraft 1.8, Minecraft 1.8.1, Minecraft 1.8.7, Minecraft 15w46a, Minecraft 1.10.2 (+63 more) |
| Fixed in | 24w44a                                                                                                          |
| Reported | 2013-03-25                                                                                                      |

## STEPS TO REPRODUCE

1. Create or enter a Minecraft world.
2. Run `/fill ~2 ~ ~1 ~2 ~6 ~15 minecraft:scaffolding`` to create a wall of scaffolding.
3. Switch to Creative mode by running `/gamemode creative``.
4. Begin flying and fly toward the scaffolding wall.
5. Fly through the scaffolding and observe whether your movement slows down while passing through it.

## OBSERVED BEHAVIOUR

When a player in Creative mode is flying and moves through a climbable block, such as scaffolding, their movement speed is reduced while they are inside or intersecting the block. This slowdown affects Creative flight even though the player is not attempting to climb the block. The player resumes normal flight speed after leaving the climbable blocks.

## EXPECTED BEHAVIOUR

While flying in Creative mode, passing through climbable blocks such as scaffolding should not reduce the player's movement speed. The player should maintain the normal Creative flight speed while entering, moving through, and exiting these blocks, just as they would when flying through non-climbable blocks. Climbable blocks should not affect movement when the player is already flying.

## ENVIRONMENT

Minecraft Java Edition Versions: 1.5.1, 1.8, 1.8.1, 1.8.7, 15w46a, 1.10.2, 16w40a, 16w41a, 1.12.1, 1.12.2, 18w20c, 1.13.1, 1.13.2, 19w12b, 19w13b, 19w14a, 19w14b, 1.14.4, 19w45a, 1.15 Pre-release 1, 1.15.2, 20w16a, 20w17a, 20w18a, 20w22a, 1.16 Pre-release 2, 1.16 Pre-release 5, 1.16 Pre-release 6, 1.16 Pre-release 7, 1.16.1, 20w27a, 20w28a, 1.16.2 Pre-release 1, 1.16.2, 1.16.3 Release Candidate 1, 1.16.3, 20w46a, 20w51a, 21w03a, 1.16.5, 21w05b, 21w06a, 21w07a, 21w08b, 21w11a, 21w15a, 21w17a, 1.17, 1.17.1, 21w43a, 1.18 Pre-release 1, 1.18, 1.18.1 Pre-release 1, 1.18.1, 1.18.2, 22w12a, 1.19, 1.19.2, 1.19.3, 1.19.4, 1.20, 1.20.1, 23w33a, 24w11a, 1.20.6, 1.21, 1.21.1, 24w36a, 1.21.2 Pre-Release 3 (includes snapshots)

## ORIGINAL DESCRIPTION

## The Bug:

Flying through climbable blocks in creative mode slows you down.

## Steps to Reproduce:

- Summon a wall of climbable blocks, for example, a wall of scaffolding by using the command provided below.

`/fill ~2 ~ ~1 ~2 ~6 ~15 minecraft:scaffolding`

- Switch into creative mode if not already and begin flying.

- Fly through the scaffolding and as you do this, pay close attention to the speed at which you travel through them.

- Take note as to whether or not flying through climbable blocks in creative mode slows you down.

## Observed Behavior:

Flying through climbable blocks in creative mode slows you down.

## Expected Behavior:

Flying through climbable blocks in creative mode would not slow you down.

## Code Analysis:

Code analysis by Avoma can be found below. An additional code analysis by \_\_null can be found in this comment. The following is based on a decompiled version of Minecraft 1.19.2 using MCP-Reborn.

net.minecraft.world.entity.LivingEntity.java

```
public abstract class LivingEntity extends Entity {
...
public boolean onClimbable() {
    if (this.isSpectator()) {
        return false;
    } else {
        BlockPos blockpos = this.blockPosition();
        BlockState blockstate = this.getFeetBlockState();
        if (blockstate.is(BlockTags.CLIMBABLE)) {
            this.lastClimbablePos = Optional.of(blockpos);
            return true;
        } else if (blockstate.getBlock() instanceof TrapDoorBlock && this.trapdoorUsableAsLadder(blockpos, blockstate)) {
            this.lastClimbablePos = Optional.of(blockpos);
            return true;
        } else {
            return false;
        }
    }
}
...

```

If we look at the above class, we can see that there is only one check that is carried out before allowing a living entity to climb a climbable block. This check is to see if the living entity is in spectator mode, and if they are, they won't be able to climb climbable blocks, and if they're not, they will be able to climb climbable blocks. The game doesn't check if the living entity is currently flying before allowing them to climb climbable blocks, therefore resulting in this problem occurring.

## Fix:

Simply adding an "if else" statement to the "if" statement to check if the living entity is a player and if they're flying before allowing

## MANUAL RESULT

|                  |                    |
|------------------|--------------------|
| Version used     |                    |
| Reproduced?      | YES / NO / PARTIAL |
| Crash file       |                    |
| Notes            |                    |
|                  |                    |
| Tested by / date |                    |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-12829>